

2(a)	1 <i>idea of:</i> add lice to container and counting lice (on trout) at death or add lice to trout (in a container) and counting lice (on trout) at death or capture trout with lice attached and count lice (on trout) at death or trout(s) with (known) different number of lice attached (in separate containers) and see which die ; 2 <i>idea of</i> repeat with container (of trout) / trout (in a container) with no lice present ; 3 named environmental condition in container maintained as constant (for, experimental / control / both, conditions) or named feature of trout / lice maintained as constant (for, experimental / control / both, conditions) ;	3
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Question	Answer	Marks										
2(b)(i)	<i>any one from:</i> location of, boat / sampling site ; (electric) current ; time of year ;	1										
2(b)(ii)	<i>any one from:</i> <i>idea that</i> young / small, trout will not be caught / AW ; ora <i>idea that</i> there will be more lice on the fish, caught / AW (as they are older / larger) ; ora	1										
2(c)(i)	allows for valid comparison; (because) number (of sea trout), sampled / AW, varies ;	2										
2(c)(ii)	paired data quote / manipulated data (to show year 1 to 2 increase) ; <i>Data quotes:</i> <table><tr><th>year</th><th>mean % with ≥ 13 attached lice</th></tr><tr><td>1</td><td>2.6</td></tr><tr><td>2</td><td>14.9</td></tr><tr><td>more / higher</td><td>12.3</td></tr><tr><td>times increased</td><td>x5.7</td></tr></table>	year	mean % with ≥ 13 attached lice	1	2.6	2	14.9	more / higher	12.3	times increased	x5.7	1
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Question	Answer	Marks
2(c)(iii)	any two from: 1 no difference between year 1 and year 2 of production cycle in 2004–2005 ; 2 number of sea trout sampled was small ; 3 (data from) only, 5 production cycles / 10 years ; 4 no statistical test carried out ;	2
2(d)(i)	there is no, relationship / correlation, between the number of lice (on trout) and the distance from the (nearest) fish farm ;	1
2(d)(ii)	(Ireland) significant / yes / ✓ / AW and (Scotland) not significant / no / ✗ / AW ;	1
2(d)(iii)	there is a negative correlation or as distance from (nearest) fish farm increases, (mean) number of lice (per trout) decreases ; ora	1

Question	Answer	Marks
2(d)(iv)	<i>Max one from MP1–3 reasons for 95% CI:</i> 1 95%, probability / confidence that / chance that, the, true / actual / population / parametric, mean lies within, this range / these limits or 2 to test whether the means (for number lice per trout at different distances from fish farms) are (significantly) different or to see / test, whether the, error bars / 95% CI / they, overlap or 3 to get a measure of how close the, calculated / AW, mean is likely to be to the, true / AW, mean or shorter, error bars / 95% CI, show true mean is likely to be closer to the calculated mean or longer, error bars / 95% CI, show true mean is likely to be further from the calculated mean ; One mark for exemplification: Exemplification of MP 2 or 3 from Fig. 2.3: <i>MP2 Exemplification:</i> <i>idea that</i> no / less overlap of (95% CI) error bars for Ireland, so (difference), is / may be, significant or <i>idea that</i> overlap of (95% CI) error bars for, Scotland / Ireland, so (difference), is not / may not be, significant ; I direct comparison between Ireland with Scotland <i>MP3 Exemplification:</i> <i>idea that</i> the, shorter / narrower, error bars for Ireland the closer the calculated mean is (likely to be) to the true mean or <i>idea that</i> the, larger / longer, error bars for Scotland, the further the calculated mean is (likely to be) from the true mean or shorter / narrower, error bars / confidence limits, with increasing distance (to nearest fish farm) in, Ireland / Scotland, shows the calculated mean is, getting / AW, closer to the true mean ; I direct comparison between Ireland with Scotland	2